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EXECUTIVE SUMMARY OF THE THESIS

TinyML-enabled Human indoor localization with UWB-radar imaging

LAUREA MAGISTRALE IN COMPUTER SCIENCE AND ENGINEERING - INGEGNERIA INFORMATICA

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1. Introduction

In the era of rapid technological evolution, the surge in embedded devices, edge computing, and IoT units is reshaping computing paradigms towards distributed and edge systems. TinyML, or Tiny Machine Learning, stands at the forefront of this transformation, by enabling AI capabilities on resource-constrained devices. Particularly crucial in IoT and edge computing environments, TinyML facilitates data processing close to the source to reduce latency, save energy, and maintain data privacy. Moreover, the integration of Ultra-Wideband (UWB) radar technology represents a cutting-edge solution that seamlessly integrates with TinyML principles. UWB radar technology leverages ultra-wideband radio waves to deliver precise sensing in compact devices while preserving privacy. These radar systems are designed for high precision in detecting environmental changes with high accuracy, for energy efficiency, with minimal power consumption, and fast data acquisition, crucial for real-time applications. The combination of advanced machine learning algorithms, through TinyML, and UWB radar technology unlocks the opportunity to develop innovative and efficient applications such as smart sensing, localization, and gesture recognition.

Between the huge number of research areas that this topic opens, this thesis explores the capabilities of UWB radars in indoor human localization task exploiting object detection algorithms.

2. Background

2.1. UWB Radar Imaging

In [2] Ultra-wideband (UWB) radar is defined as a wireless technology that transmits sub-nanosecond pulses, using very low power radio waves to communicate over short distances. It is characterized by the use of a broad spectrum of frequencies, between 3.1 GHz and 10.6 GHz, and a bandwidth typically greater than 500MHz or at least 20% of the radar center frequency, combined with very short narrow pulses. It enables high range resolution and imaging capability, where range resolution is the ability to distinguish between two closely spaced objects. With multiple receiver antennas, a radar pulse reflecting off a target may arrive at one of the antennas later than the other. This time difference makes it possible to calculate the angle the reflection is coming from, meaning that we can combine the distance information, provided by each receiver, with the angular information, to get an understanding of where the reflection is coming from in 2D-space. A Single-

Input Multiple-Output (SIMO) UWB radar system can be model as a Uniform Linear Array (ULA) of sensors, representing a collection of N_{RX} receiver antennas physically deployed on the UWB device, and equally spaced of δ_{RX} along a straight line, where transmitter antenna position is approximated with the array's first element position.

2.1.1 Digital Beamforming

Digital Beamforming (DBF) is a time-domain signal processing technique used to improve the signal-to-noise ratio of the signal from a specific direction, or Digital Beam, by delaying and summing the signals from an ULA of sensors. Digital beams are generated by calculating phase-shifts (i.e. delays), that will focus a digital beam to a given angle, for all the receiving antennas taking as reference the first element of the array. These delays can then be applied as weights to the ULA receivers and the resulting weighted signals can be summed together to form a digital beam at the given angle. More formally, having a list of angles $\Theta = \{\theta_1, \dots, \theta_{N_a}\}$ called Steering Vector, then we can compute the DBF weights $\forall \theta_i \in \Theta, \forall n_{RX} \in \{1, \dots, N_{RX}\}$ as:

$$W(\theta_i) = \exp \left[-j \frac{2\pi}{\lambda} \delta_{RX} (n_{RX} - 1) \sin \theta_i \right] \quad (1)$$

where λ is the wavelength of the transmitted radar pulse, δ_{RX} is the distance between the ULA receiver antennas and j is the imaginary unit. The result is the matrix $W \in \mathbb{C}^{N_a \times N_{RX}}$. Let $s_n \in \mathbb{C}^K$ be the complex signal received by the n_{RX} -th element of the array in a single radar frame, $n_{RX} \in \{1, \dots, N_{RX}\}$, where $K \in \mathbb{N}$ is the number of quantized distances in the radar acquisition range, also called Range profile. The DBF algorithm takes as input the set of radar acquisitions from N_{RX} antennas, $S \in \mathbb{C}^{N_{RX} \times K}$, and maps them to a Range-Angle map $S_{angle} \in \mathbb{R}^{N_a \times K}$, which contains the magnitudes of the acquired signals in the angle directions $\Theta = \{\theta_1, \dots, \theta_{N_a}\}$ and range profile K , for a single radar frame.

2.1.2 MIMO radar

One of the most promising developments in UWB radar technology is the integration of

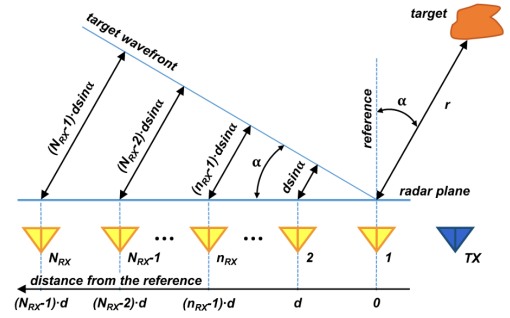


Figure 1: ULA of receiver antennas [4]

Multiple-Input Multiple-Output (MIMO) systems. By leveraging the spatial diversity of antenna configurations, multiple transmitters of MIMO UWB radars can provide more detailed spatial information about the environment and to improve imaging and localization capabilities in more dimensions (Distance, Azimuth, Elevation). In the context of DBF, it is possible to increase the angular resolution by increasing the number of receivers in the ULA without deploying more physical antennas, using the phase differences at the different locations of the transmitter and receiver [1][7]. In particular, a ULA of N_{RX} receivers, equally spaced of a distance δ_{RX} , can be extended to a virtual ULA (vULA) of $N_{RX}^V = N_{TX} \times N_{RX}$ antennas, where N_{TX} is the number of antennas in the ULA of transmitters with a spacing of $\delta_{TX} = \delta_{RX} \times N_{RX}$. Nevertheless, more antennas means more energy consumption and we will evaluate the performance of 2D UWB radar imaging with different antennas configurations.

2.2. Object Detection

Object detection plays a pivotal role in the field of computer vision, facilitating the identification of objects within images or videos by determining their spatial location and boundaries and by categorizing these objects into distinct classes. Deep learning object detection networks can be trained for a wide range of tasks such as people counting, pedestrian recognition, face detection etc.

Let X be an image, or a video frame, and let $\Omega = \{(y_1, b_1), \dots, (y_L, b_L)\}$ be a set of annotations, where (y_l, b_l) represents the class and the position of the l -th object within the image. The position b is composed of 4 values, $(x_{min}, x_{max}, y_{min}, y_{max})$, which represents the

portion of image in which the object can be found, called Bounding Box.

Taking X as input, an Object Detector must be able to generate a set $\Omega^{pred} = \{(y_1^{pred}, b_1^{pred}), \dots, (y_L^{pred}, b_L^{pred})\}$ of predictions where (y_l^{pred}, b_l^{pred}) represents the class and the Bounding Box of the l -th predicted object.

The quality of the predictions is evaluated by a scoring function $f(\Omega^{pred}, \Omega) = \{\sigma_1, \dots, \sigma_L\}$ where σ_l is the score of the l -th predicted object. The main disadvantage of object detectors is that they are computationally very expensive and require a huge amount of annotated training data. One of the most recent advances in TinyML is the execution of object detection algorithms on resource constrained devices exploiting compression techniques and network re-design.

Faster Objects, More Objects (FOMO) is an example of novel lightweight deep learning object detection model provided by Edge Impulse [3]. The efficiency of FOMO takes advantage from low computational power required by simple image classification adding localization and object count information. Differently from standard object detection algorithms, taking an input image X , FOMO is able to generate a set $\Omega^{pred} = \{(y_1^{pred}, C_1^{pred}), \dots, (y_L^{pred}, C_L^{pred})\}$ of annotations where (y_l^{pred}, C_l^{pred}) represents the class and the Centroid, i.e. the Bounding Box center of the l -th predicted object, and L is the number of objects. It exploits Transfer Learning, using early layers of MobileNetV2 Features Extractor (FE) pre-trained on Image Classification tasks, to extract diffused lower-resolution image of the input, preserving its locality, that is then processed through N_{Conv} 2D convolutional layers of N_{Filt} 1×1 filters.

Let $N_{Class} + 1$ be the number of detected object classes plus background, then the final 2D convolutional layer of FOMO produces $N_{Class} + 1$ per-region probability maps of size (H, W) , where a cell $m_n[i, j]$ of the n -th map represents the probability that the corresponding position in the input image X belongs to the n -th object class, such that:

$$\sum_{n=1}^{N_{Class}+1} m_n[i, j] = 1 \quad (2)$$

$$\forall i \in \{1, \dots, H\}, \forall j \in \{1, \dots, W\}.$$



Figure 2: Example of bees camera-based detection with FOMO, Red dots are detected bees' centroids[3].

Then highlighted areas in the n -th map, i.e. adjacent cells with probability $m_n[i, j]$ higher than a confidence threshold τ_c , represents areas in which the objects belonging to the n -th class can be found and from them is computed the (i, j) position of their centroids.

The *MobileNetV2* FE can be configured with different hyper-parameters such as α , that controls the number of filters inside layers of the FE, and the cut-point P_C , that is the output layer of the FE which controls the resolution (H, W) of the probability maps, setting also a trade-off between memory size and object size detection capability of the model.

FOMO is designed for camera-based object detection on constrained devices leveraging on a very lightweight architecture and post training quantization. In addition, training it on centroids of objects, makes it suitable for applications where it is not necessary to know the size of the object but its position is enough, like Figure 2.

3. Related Works

One of the most relevant research area in UWB technologies is the human localization in sectors like automotive and indoor environments, as also highlighted by [2]. For example, Authors of [4] proposes a signal processing algorithm using a millimeter-wave Single-Input Multiple-Output (SIMO) Frequency Modulated Continuous Wave (FMCW) radar to automatically determine the number and 2D positions (azimuth angle and range) of multiple stationary human subjects. It aims to isolate the Doppler signals caused by each subject's vital signs (car-

diopulmonary activity) from static clutter and multipath reflections using Fast-Fourier Transform and Digital Beamforming to obtain Angle of Arrival of signals. The number of subjects is determined by counting the number of extracted vital signs signatures. 2D localization is performed by measuring the distance to the subjects' chests where the vital signs are sensed. [7] presents a complete signal processing pipeline for multi-person detection, localization and vital sign estimation using a stepped-frequency continuous wave (SFCW) UWB radar in degraded visibility conditions. Key steps includes clutter removal, virtual antenna array synthesis, 2D MUSIC localization algorithm, model order estimation for number of persons, and spatial filtering to extract vital signs like breathing rate. These works achieve state-of-art results in human localization, exploiting complex signal processing techniques which do not take into account resource limitations for on device localization tasks. There are few works which combine machine learning to UWB radars technology and bring their power on tiny device. An example is [5] in which is introduced a novel TinyML solution for presence detection using Ultrawideband (UWB) radar data through a lightweight pre-processing pipeline, and a family of tiny convolutional neural networks (TyCNNs). [6], instead, proposes using UWB radar and TyCNNs to develop privacy-preserving person counting and object detection solutions for tiny devices. Nevertheless, no existing solution explores the usage of tinyML in 2D human indoor localization, even less using MIMO UWB radar imaging. Previous works, in fact, requires the collection of long windows of data while the proposed solution in this thesis works instantaneously.

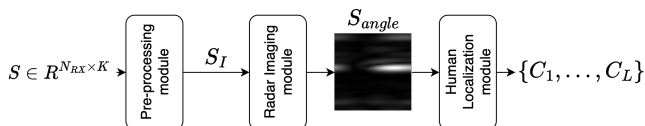


Figure 3: Indoor human 2D-Localization Pipeline

4. Problem Setting & Proposed Solution

The main purpose of this thesis is to develop a pipeline for 2D (Range-Azimuth) indoor human localization through UWB radar signals and evaluate its memory and computation efficiency for a constrained device.

Using the same formalization introduced in *Section 2.1*, the pipeline must take in input the set S of radar acquisitions from N_{RX}^V antennas of a vULA, and output the number L and the Range-Angle(Azimuth) positions C_l of Human subjects within the scanned environment. The proposed solution is composed of 3 sub-modules:

- a *pre-processing module*;
- a *MIMO UWB radar imaging module* to map the set S to a Range-Angle map S_{angle} , characterized by a Range profile K and a vector of Angles Θ ;
- the *Human Localization on Radar Maps (HLRM) module*, to obtain from S_{angle} the number L and the positions $C_l = (r_l, \theta_l)$ of human targets, where (r_l, θ_l) are the Range and Azimuth coordinates within the input Range-Angle map.

The proposed Indoor human 2D-Localization Pipeline is depicted in Figure 3. The radar data S from N_{RX}^V antennas, are raw complex signals containing noise from static clutter reflections, so we have to "clean" the signal with a low-pass filter, which preserve human targets' vital signs, before they can be processed by the MIMO radar imaging module. For this reason, the *pre-processing module* exploits a moving average filter, as described in [7], and outputs the cleaned S_I . Then the *MIMO radar imaging module* takes in input the set of pre-processed radar acquisition $S_I \in \mathbb{C}^{N_{RX}^V \times K}$ from N_{RX}^V antennas and map it to a Range-Angle map $S_{angle} \in \mathbb{R}^{N_a \times K}$ using the weights $W(\theta_i)$ of formula (1) in the following:

$$S_{angle}[i, :] = S[:, :] \times W(\theta_i) \quad (3)$$

$\forall i \in \{1, \dots, N_a\}$. An example of Range-Angle map can be seen in Figure 4.

The *HLRM module* adapts the FOMO [3] algorithm to work on radar maps, instead of the images it was designed to work on. The algorithm is well fit for the task, since Human targets are just a kind of "globe" within UWB

radar Range-Angle images, and we just need to localize and count them. The formalization is very similar to the one presented in *Section 2.2* with the difference that in our solution the input is a Range-Angle radar map S_{angle} , containing $\Omega = \{(y_1, b_1), \dots, (y_L, b_L)\}$ human targets, and the output is the set $\Omega^{pred} = \{(y_1^{pred}, C_1^{pred}), \dots, (y_L^{pred}, C_L^{pred})\}$ of L detected humans and their centroid positions $C_l = (r_l, \theta_l)$. In particular, we want to detect only objects of class *Human*. Consequently, the number of probability maps produced by the *HLRM module* last convolutional layer is $N_{Class} + 1 = 2$, considering also the *Background* class. The *HLRM module* architecture relies on the following parameters:

- the *object_weight* w_{obj} parameter can be used to balance situations in which the objects to detect are very few compared to the background;
- α controls the number of filters inside layers of the FE; cut-point P_C , that is the output layer of the FE;
- N_{Conv} 2D convolutional layers before the output one;
- and the number of filters N_{Filt} inside them.

The algorithm is trained with an optimizer Opt and a learning rate l_r . The human localization on radar maps module has to be runnable on constrained devices, so it has to respect memory and execution limits given by the device used. Therefore, this research aims to evaluate the detection capabilities of the proposed pipeline with Range-Angle maps generated from different antenna configurations, considering that each physically deployed antenna leads to more energy consumption. *Section 5* contains experiments for different UWB devices with different number of antennas used. Each configura-

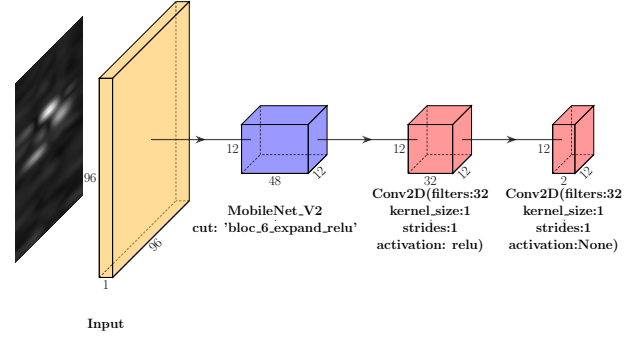


Figure 5: *HLRM module* baseline architecture with $P_C = \text{block}_6$, $\alpha = 0.35$, $N_{Conv} = 1$, $N_{filt}=32$ and $w_{obj} = 100$.

tion is used to train and to test *HLRM module* with the hyper-parameters previously described to find the best ones that maximize its performance in human indoor 2D-Localization, starting from the Baseline model in Figure 5.

5. Experimental setting

This section provide an overview on the experiments conducted to evaluate our proposed solution, using the formalization in *Section 2*.

To train and then test the performance of our Pipeline, we used 2 datasets composed of radar scans of an indoor environment, setting 3 vULA configurations for our experiments:

- **8-RX** and **4-RX** vULAs with Multi-Person Localization and Vital Sign Estimation Radar Dataset [7] collected with Vayyar's SFCW MIMO UWB radar, *Section 5.1* ;
- **3-RX** vULA dataset collected with Novelda X7 MIMO UWB Radar, *Section 5.2*.

Each dataset is composed by a set of M radar scans $S \in \mathbb{C}^{N_{RX} \times K}$ for each N_{TX} of the MIMO radar. Then the *MIMO UWB radar imaging module* have been used to obtain a dataset of M Range-Angle maps $S_{angle} \in \mathbb{R}^{N_a \times K}$ for the relative antennas configuration. Each scenario is splitted into 80% for training and 20% for testing purposes, and values of each S_{angle} is normalized in the range $[0; 1]$, before the model training phase.

5.1. Multi-Person Localization Radar Dataset

This dataset is collected by authors of [7] with a SFCW MIMO UWB radar, by manufacturer Vayyar, for *through-wall vital signs estimation*

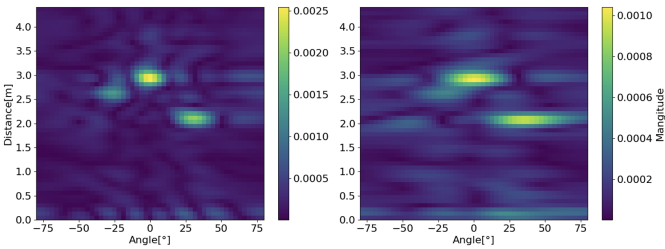


Figure 4: Range-Angle radar maps of 8RX(*Left*) and 4RX(*Right*) vULAs, with 3 Human Targets at distance and angle $(2m, +30^\circ)$, $(2.5m, -30^\circ)$ and $(3m, 0^\circ)$

through UWB radar purpose. The radar operates at central frequency f_c of 7.15GHz with a bandwidth B of 1.7GHz, and is composed by 18 antennas disposed in a grid with an inter-element spacing $\delta_{RX} = 2\text{cm}$, as shown in Figure 6, each one sampling a signal of $K = 137$ range bins, with a range resolution of 8.8cm. It contains 62 scenarios, sampled at a frequency $f_s = 10\text{Hz}$, with up to five static persons captured in different postures, angles and ranges including obstacles that block the radar line of sight. A subset of 14 scenarios for a total of $M = 13200$ radar scans, have been selected for our experiments, removing ones which comprise obstacles and human targets too close to be theoretically sensed by the UWB radar system as 2 separated targets. Using only scans from the 2TX and 4RX antennas highlighted in Figure 6, we are able to synthesize a vULA of $N_{RX}^V = 8$ and a vULA of $N_{RX}^V = 4$ to be processed by the *MIMO UWB radar imaging module*. Then we obtain a Range-Angle map dataset for each antenna configuration, with $M = 13200$, $N_a = 51$ angles in range $[-80^\circ; +80^\circ]$, and $K = 51$ range bins up to 4.5 meters. A bounding box b_l is generated around each y_l human target in each map using meta data information on target positions provided by [7].

5.2. Novelda X7 dataset

The X7, by manufacturer Novelda, is a MIMO UWB impulse radar provided of only 2 transceiver antennas, $N_{RX} = N_{TX} = 2$, designed for robust low power presence detection and beamforming applications. A transceiver antenna can act as both transmitter and receiver in a synchronized way, allowing to synthesize a vULA of $N_{RX}^V = 3$, since $\delta_{TX} = \delta_{RX}$ since, with inter-element spacing $\delta_{RX} = 2\text{cm}$. This radar

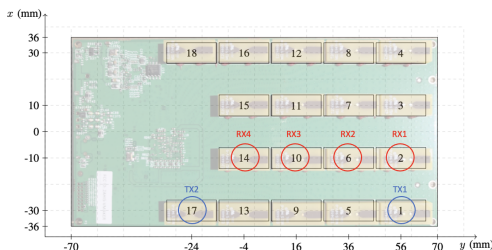


Figure 6: Walabot antennas disposition [7], highlighting TX(Blue) and RX(Red) antennas used to build the vULA used for this work.

RX	P_C	α	l_r	w_{obj}	N_{conv}	N_{filt}
8	$block_6$	0.35	0.01	50	1	64
4	$block_6$	0.35	0.01	50	2	32
3	$block_6$	0.35	0.001	100	2	32

Table 1: Best *HLRM module* hyper-parameters for each antennas configuration setting.

operates at a center frequency $f_c = 7.875\text{GHz}$, a bandwidth $B = 750\text{MHz}$, and a range profile of $K = 192$ bins, with a range resolution of 7.14cm. We used it to collect radar scans from only 2 physically deployed antennas, of a simpler environment with a single static human target (y, C) , located in 9 different combinations of distances $\{1\text{m}, 2\text{m}, 3\text{m}\}$ and angles $\{-35^\circ, 0^\circ, +35^\circ\}$, for a total of $M = 9000$ radar scans. Then using *MIMO UWB radar imaging module* we obtain the Range-Angle map dataset for the $N_{RX}^V = 3$ vULA antennas configuration, with $M = 9000$, $N_a = 51$ angles in $[-60^\circ; +60^\circ]$, and $K = 51$ range bins up to 3.6 meters. A bounding box b_l is generated around the y_l human target in each map using the relative experimental position.

5.3. Experiments

The experiments were conducted to evaluate the performance of our Pipeline in **8-RX**, **4-RX** and **3-RX** vULAs settings, from respectively 6, 5 and 2 physically deployed antennas. For each setting, we evaluated visually the performance of the *MIMO UWB radar imaging module*. The results of imaging process can be seen in Figure 4. Furthermore, we evaluated the effects on the various hyperparameters of the *HLRM module* algorithm on a validation dataset (composed of 20% of the data of the training set that were excluded from the training). We used this validation process to select the best hyperparameters for this algorithm, which can be seen in Table 1, for the 3 different settings respectively. We evaluated the validation $F1 - score \in [0; 1]$. This metric is useful in applications where both the number and the quality of accurate predictions are important, assessing the model's ability to correctly identify objects while minimizing the number of false positives and false negatives. The training process is performed with `EARLYSTOPPING` to reduce overfitting,

8RX	Memory	Exec. Time	Test Acc.
HLRM	1MB	3ms	92.8%
YoloV5	3.4MB	26ms	85.5%

4RX	Memory	Exec. Time	Test Acc.
HLRM	993KB	3ms	83.8%
YoloV5	3.4MB	26ms	25.4%

Table 2: Localization results on 8RX and 4RX experimental datasets.

REDUCELRONPLATEAU and $Opt = Adam$ trying $l_r \in \{0.01, 0.001, 0.0001\}$.

In Section 6 is resumed the performance of the best model for each antennas configuration setting, tested on the relative testing dataset. The performance of tested models is evaluated on *Accuracy* score, defined as the percentage of testing radar scans with Precision $P > 80\%$, where $P = \frac{\text{true positive detected humans}}{\text{all detected humans}}$ within a radar scan S . The focus on Precision and setting a threshold highlights the importance of not just detecting humans, but detecting them accurately with minimal false positives. *Accuracy* is used to compare object detection performance of models that uses different metrics on predicted bounding boxes, like *YoloV5*.

6. Results

Now we present the results of the experiments on our Human Indoor 2D-Localization Pipeline with different antennas configuration. The test Accuracy of our models is compared to an EdgeML version of *YoloV5*, model size of 3.4MB, trained and tested on same datasets of our experimental settings. We evaluated also the inference time and the memory (flash + RAM) usage on a virtual environment simulating a RaspberryPi4, of both *YoloV5* and the proposed solution.

Table 2 highlight a very good performance for the **8-RX** setting, which degrades in the **4-RX** as we expected since with less antennas the UWB sensing capability is reduced resulting in more noisy Range-Angle maps and then a reduced human localization performance.

Moreover the *HLRM module* models outperforms *Yolov5* models in accuracy, memory and execution time, demonstrating to be a very lightweight, high performing, detection models

also in a more constrained setting as the **3-RX** one, with only 2 physically deployed antennas, less bandwidth and a single target to detect, Table 3. The large difference in results, may be also due to the relatively limited dimensions of the training datasets: it is possible that *YoloV5*, being a large architecture requires a larger amount of training data in order to be trained appropriately.

3RX	Memory	Exec. Time	Test Acc.
HLRM	993KB	3ms	98.4%
YoloV5	3.4MB	26ms	83.3%

Table 3: Localization results on 3RX experimental dataset.

7. Conclusions

This thesis demonstrates the effectiveness of combining UWB technology and TinyML algorithm for human indoor localization through radar maps in a controlled environment. We showed how combining different antennas configurations it is possible to obtain different localization performance, making our solution suitable for various applications depending on the desired *memory footprint/energy consumption* trade-off. Future directions include the porting of our solution on a tiny device to test its real-time capabilities on a physical device. The most challenging part of this work is to find a dataset that is feasible for training our solution, due to the lack of research on the topic. It may then be useful to extend the dataset collected in this work with more complex scenarios to make the solution more robust in different environments. Finally, UWB radar maps provided a valuable visual representation of the analyzed environment, then optimizing or modifying the TinyML architecture it could be possible to extend the localization with tracking, or to extract information other than the number and positions of targets, for task such as 2D gesture recognition.

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